

Kiosk Deployment & 24/7 Ops Guide

Infrastructure replication for Gen2/Gen3 Ops Consoles on Windows 11 MiniPCs.

0. Critical Power & Sleep Management

To prevent "Modern Sleep" from suspending the Python process, these changes are mandatory:

- **Registry Override:** Navigate to `HKLM\System\CurrentControlSet\Control\Power`. Create a DWORD (32-bit) named `PlatformAoAcOverride` and set it to `0`.
- **USB Selective Suspend:** Open Control Panel > Power Options > Change plan settings > Change advanced power settings. Set **USB settings** > **USB selective suspend setting** to `Disabled`.
- **Windows UI:** Set Screen and Sleep timers to **Never**.

See [Appendix A](#) for a code-based fail-safe if Windows Updates revert these settings.

1. Hardware & OS Level Auto-Boot

Beelink BIOS Configuration

Repeatedly tap `Del` on boot. Navigate to **Advanced** > **Power Management**. Set **Restore AC Power Loss** to `Power On` (or `S0 State`). Press `F4` to Save.

Windows 11 Autologon

1. In Registry, set `DevicePasswordLessBuildVersion` to `0` at `HKLM\Software\Microsoft\Windows NT\CurrentVersion>PasswordLess\Device`.
2. Run `netplwiz`, uncheck "Users must enter a password", and provide the account credentials.

2. VS Code Profile & Environment

- Create profile `AutoDebug` (Badge: **AU**).
- Install extensions: **Python** and **Auto Run Command**.
- In Profile Settings (JSON), add the startup trigger:

```
{ "auto-run-command.rules": [{ "command": "workbench.action.debug.start", "event": "onStartupFinished" }] }
```

3. Launch Configuration & Persistence

In `launch.json`, ensure your `args` are defined for the specific hardware environment:

```
{
  "name": "Gen20psConsole_Debug",
  "type": "debugpy",
  "request": "launch",
  "program": "${workspaceFolder}/MasterControl.py",
  "args": ["--port", "COM3", "--mode", "full"]
}
```

Place a shortcut in `shell:startup` pointing to the workspace with the `--profile "AutoDebug"` flag.

Appendix A: Python "Stay Awake" Fail-safe

Add this to the top of `MasterControl.py`. This uses Windows APIs to prevent sleep regardless of Registry settings.

```
import ctypes
# 0x80000000 = Continuous, 0x00000001 = System Required
def prevent_sleep():
    ctypes.windll.kernel32.SetThreadExecutionState(0x80000000 | 0x00000001)

prevent_sleep()
```

Appendix B: Device-Level USB Power Management

If selective suspend is still causing issues for specific equipment, perform the "Paranoia Check":

1. Right-click Start > **Device Manager**.
2. Expand **Universal Serial Bus controllers**.
3. Right-click each **USB Root Hub** > **Properties**.
4. Under **Power Management** tab, uncheck "Allow the computer to turn off this device to save power."